

Smart Transformer Oil Temperature and Load Monitoring System using ESP32

Real-Time Transformer Health Monitoring with Fault Detection and Store-and-Forward Mechanism



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PROBLEM STATEMENT



Power transformers operate continuously under varying load conditions.

Excessive oil temperature and overloading can lead to



Reduced transformer life



Insulation degradation



Unexpected failures



High maintenance cost



Traditional monitoring methods require manual inspection and cannot provide continuous real-time monitoring.



PROPOSED SOLUTION

This project develops an **ESP32-based** monitoring system capable of



Monitoring transformer **oil temperature**



Monitoring transformer **load current**



Classifying transformer **health**



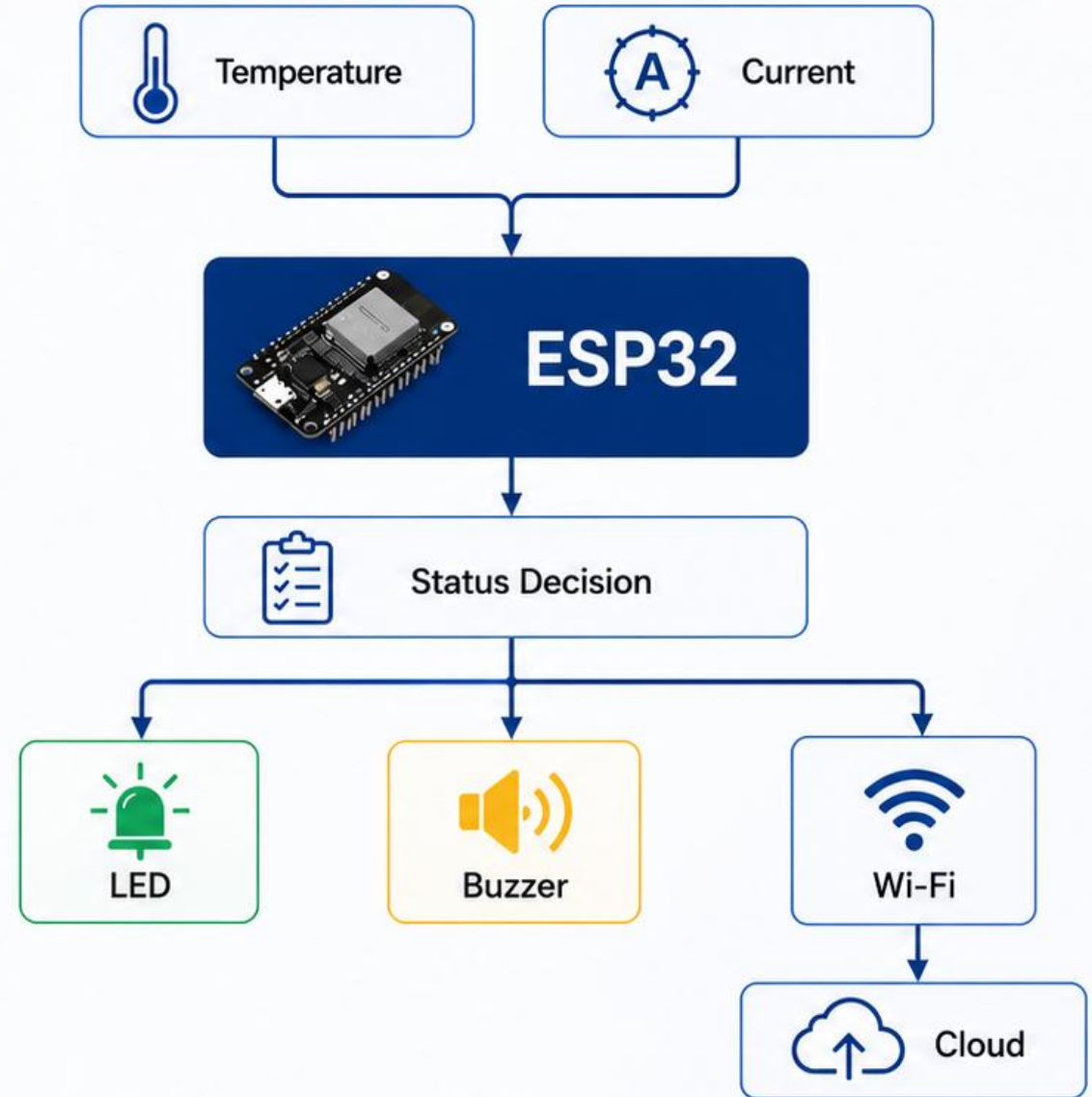
Providing local alerts using **LEDs and buzzer**



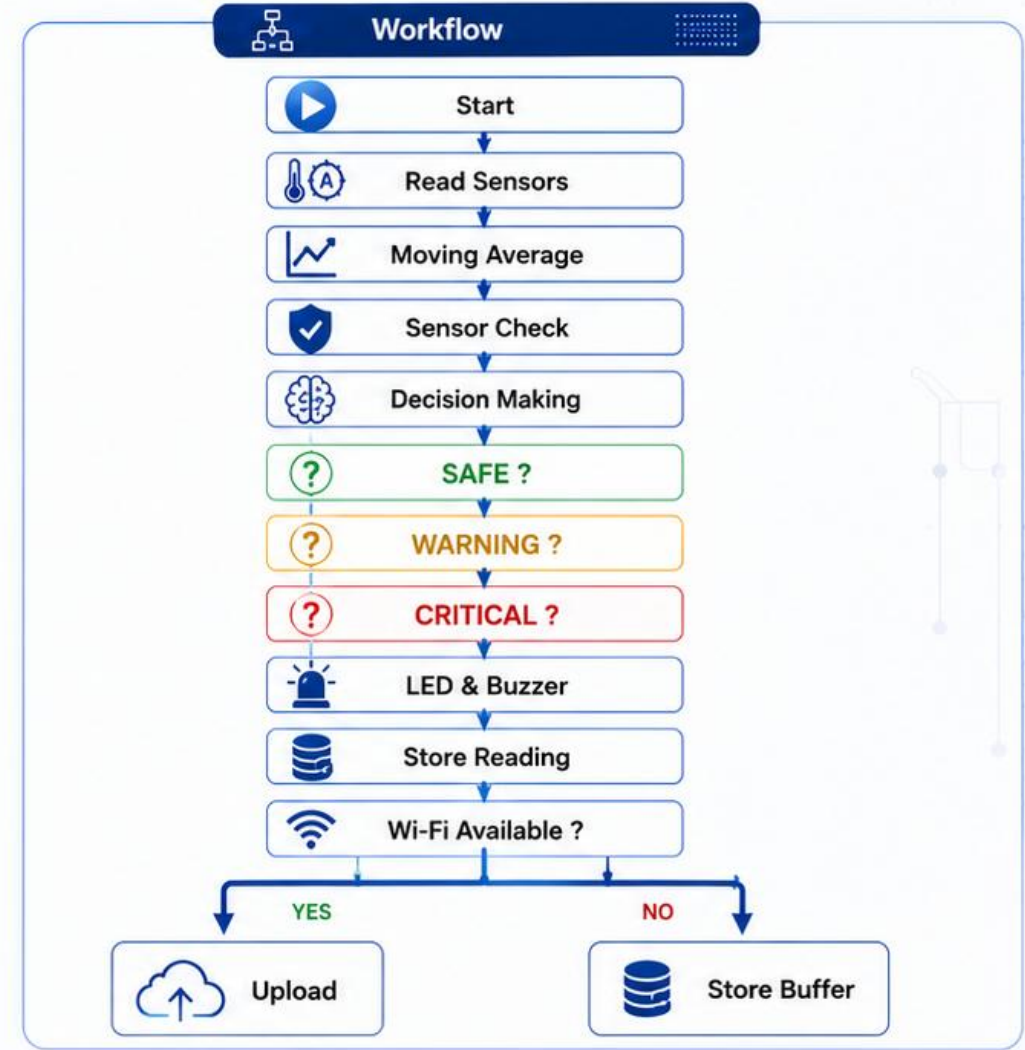
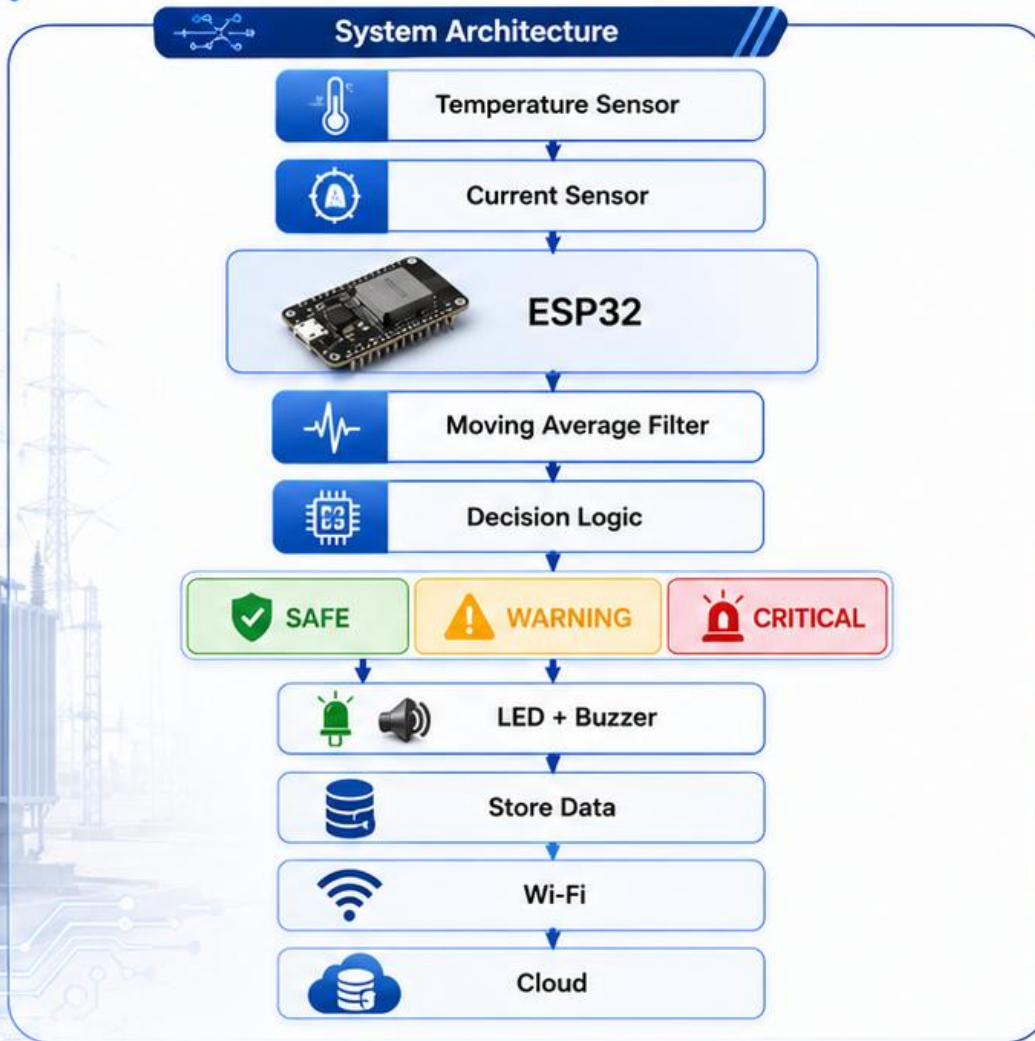
Buffering readings during network failure



Uploading stored readings once Wi-Fi is restored



SYSTEM ARCHITECTURE & WORKFLOW





TASK 1 | DATA ACQUISITION

► Read Temperature and Load Current



Current implementation

- Simulation uses potentiometers connected to ESP32 analog pins.

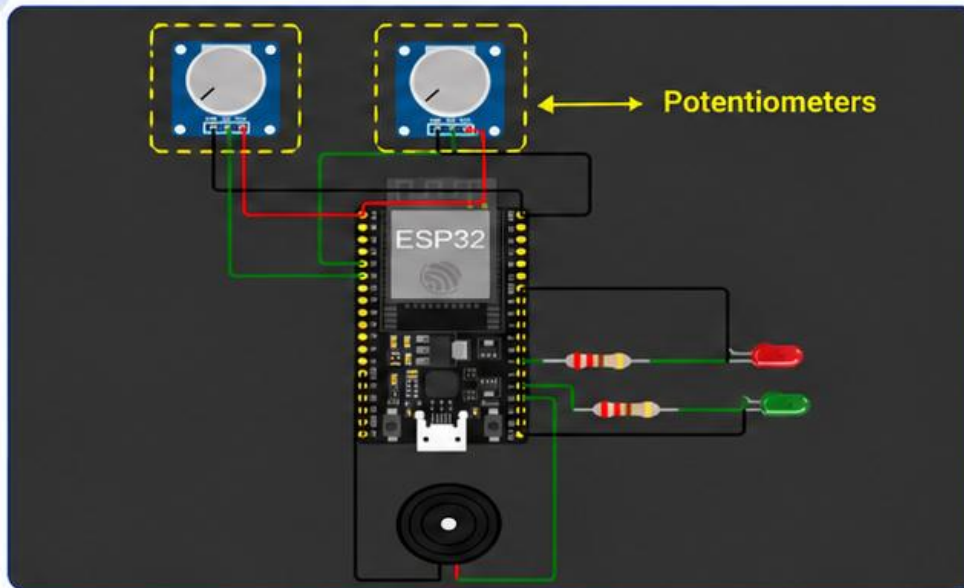


Future implementation

- LM35 / DS18B20
- ACS712 Current Sensor



Wokwi Simulation Screenshot



TASK 2 | SENSOR VALIDATION

► Plausibility Check



Purpose

Reject impossible sensor values.



Example

Temperature			
Valid	20°C	80°C	110°C
Invalid	-40°C		350°C

Current			
Valid	20A	45A	70A
Invalid	600A		



TASK 3 – ALERT SYSTEM



Alert Conditions



SAFE

● Green LED ON

- Temperature and Current within safe limits
- ESP32 displays SAFE
- Data stored continuously



WARNING

● Yellow LED ON

- Parameters approaching threshold
- Consecutive validation before alert
- Preventive maintenance recommended



CRITICAL

● Red LED ON ● Buzzer ON

- Temperature/Current exceed limit
- Reading stored for cloud upload



Wokwi Demonstration

SAFE STATE



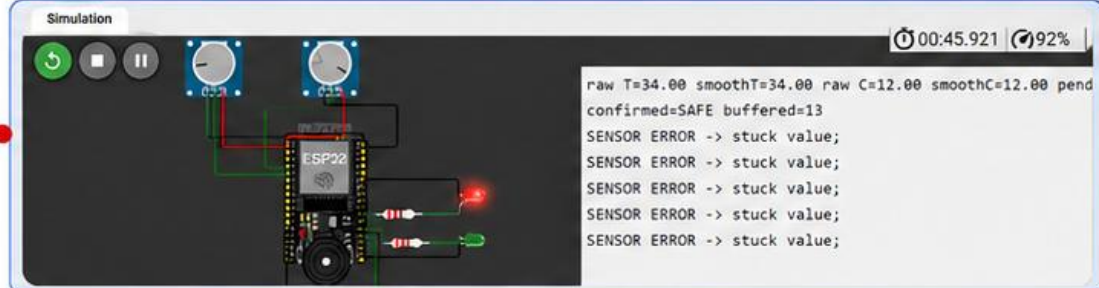
Transformer
Operating Normally



SENSOR ERROR



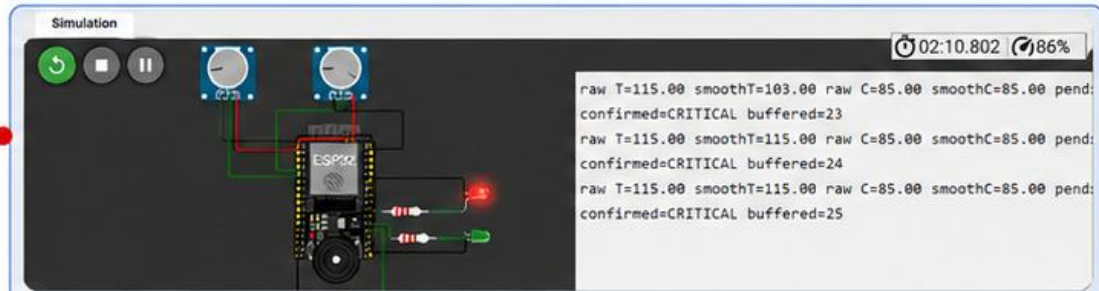
Sensor Stuck
Detected



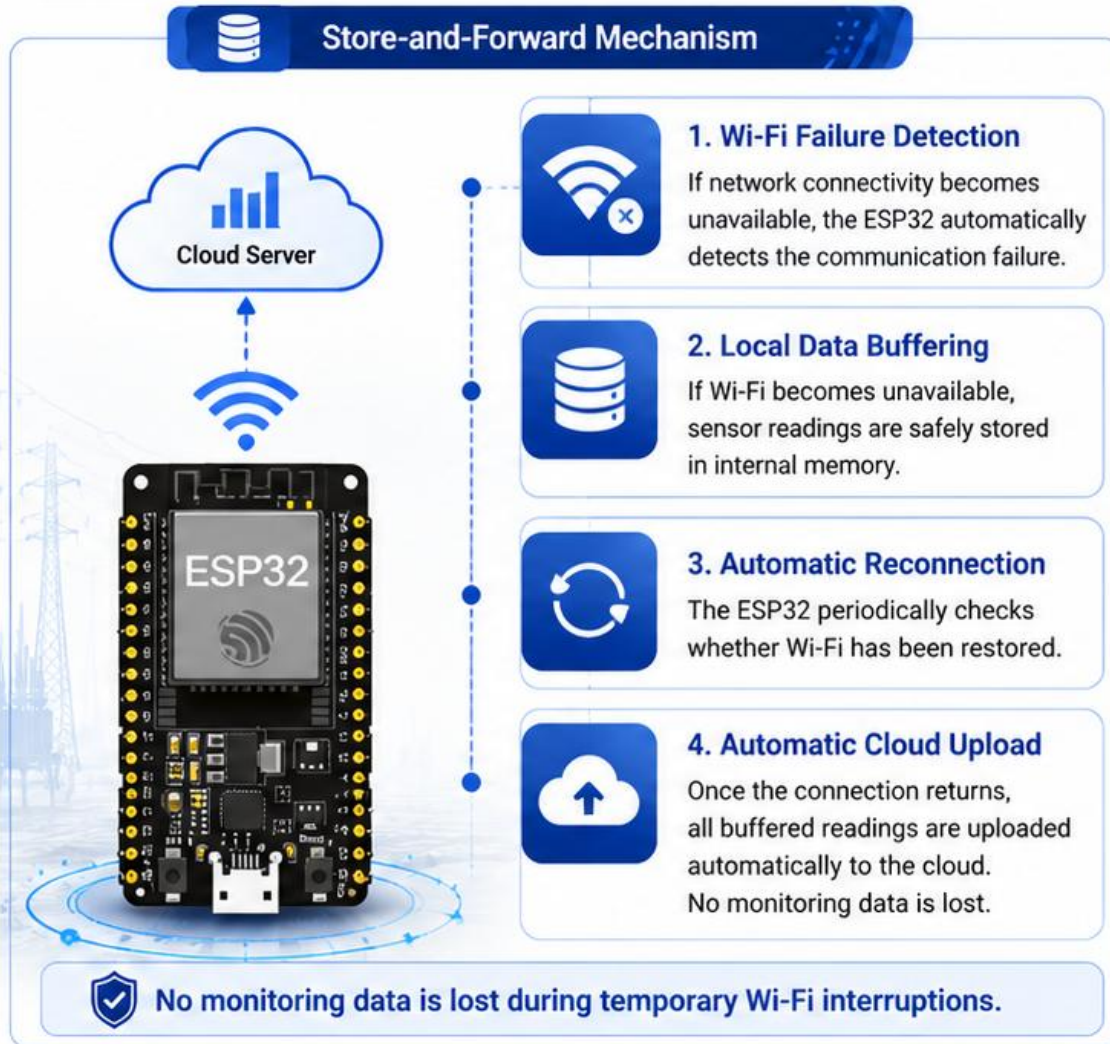
CRITICAL STATE



Over Threshold
Condition



TASK 4 – STORE-AND-FORWARD MECHANISM



CONCLUSION

The developed **ESP32-based** monitoring system demonstrates a practical approach to transformer health monitoring.

The system successfully:

-  Monitors **temperature** and load
-  Detects **abnormal** conditions
-  Generates local **alarms**
-  Handles **sensor failures**
-  Buffers **data** during communication loss
-  Provides a foundation for future **IoT-based** transformer monitoring



THANK YOU

? **Any Questions?**



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